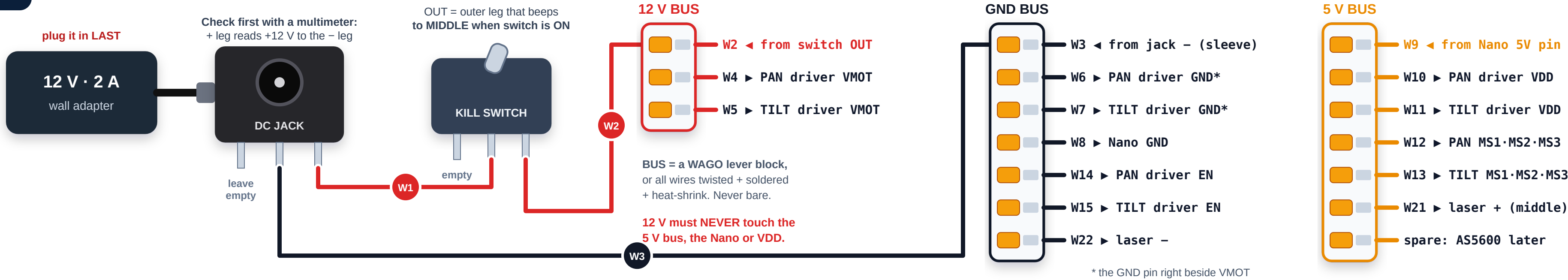


ZeroDrift MK2 · wiring WITHOUT a breadboard

Every wire has a number (W1–W22). Wire with EVERYTHING unplugged. Tick each wire in section ③ when it is done.

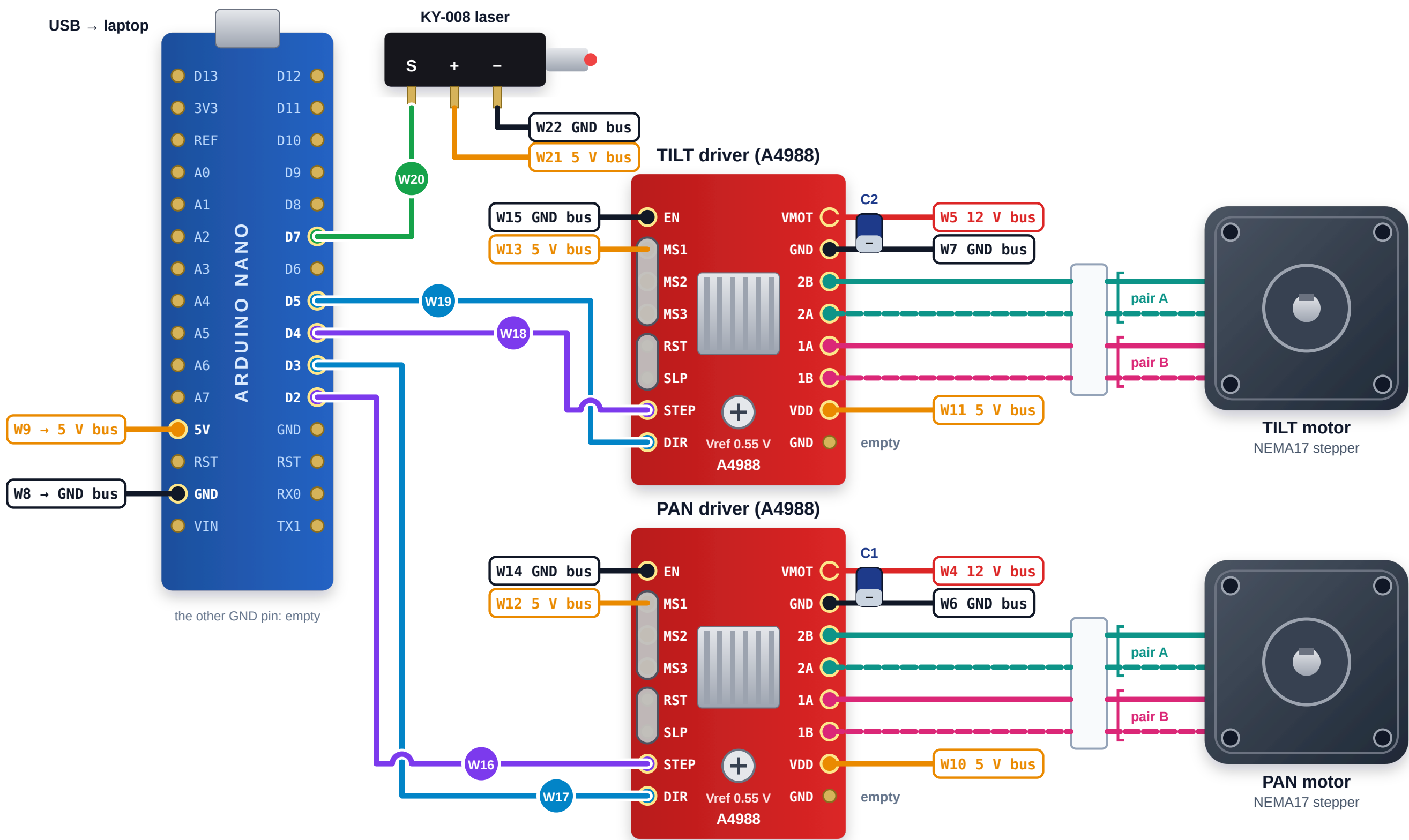
pins match rig_firmware_v2.ino
pan D2/D3 · tilt D4/D5 · laser D7

1 Power 12 V only to the drivers · 5 V from the Nano · one shared GND



2 Nano → drivers → motors

Drawn from the top. Pin names are also printed on each board: match them.



⚠ Solder bridges (each driver)

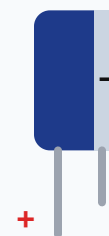
- MS1 + MS2 + MS3 joined with one blob, then ONE wire to 5 V → 1/16 microstep
- RST + SLP joined with one blob (no wire)
- EN gets ONE wire to GND



⚡ Capacitor C1 / C2 (100 µF, 25 V+)

Solder it right at the driver pins:

- long leg (+) → VMOT
 - striped leg (–) → the GND beside VMOT
- Backwards = it can burst. Check twice.



🔌 Find the motor coil pairs

Multimeter on Ω. Two wires that read 2–4 Ω are ONE coil = one pair.

- pair A → 2B + 2A • pair B → 1A + 1B
- Motor turns the wrong way? Swap the 2 wires of ONE pair. Never unplug with 12 V ON.

🎨 Wire colours in this drawing

- | | |
|--------------------|-------------------|
| 12 V (motor power) | GND (0 V) |
| 5 V (logic) | STEP signal |
| DIR signal | laser signal |
| motor coil pair A | motor coil pair B |

3 Tick every wire

Power OFF while wiring. Beep-test 12 V bus → 5 V bus: must be SILENT before power-up.

- | | | |
|--|--|---|
| ☐ W1 jack + pin → switch MIDDLE leg | ☐ W11 5 V bus → TILT VDD | ☐ W21 5 V bus → laser + (middle) |
| ☐ W2 switch OUT leg → 12 V bus | ☐ W12 5 V bus → PAN MS1-3 blob | ☐ W22 GND bus → laser – |
| ☐ W3 jack – sleeve → GND bus | ☐ W13 5 V bus → TILT MS1-3 blob | ☐ C1 100 µF at PAN: + VMOT, – GND |
| ☐ W4 12 V bus → PAN VMOT | ☐ W14 GND bus → PAN EN | ☐ C2 100 µF at TILT: + VMOT, – GND |
| ☐ W5 12 V bus → TILT VMOT | ☐ W15 GND bus → TILT EN | ☐ M each motor: pair A → 2B/2A |
| ☐ W6 PAN GND (by VMOT) → GND bus | ☐ W16 Nano D2 → PAN STEP | ☐ M each motor: pair B → 1A/1B |
| ☐ W7 TILT GND (by VMOT) → GND bus | ☐ W17 Nano D3 → PAN DIR | ☐ B both drivers: MS1-2-3 blob |
| ☐ W8 Nano GND → GND bus | ☐ W18 Nano D4 → TILT STEP | ☐ B both drivers: RST-SLP blob |
| ☐ W9 Nano 5V → 5 V bus | ☐ W19 Nano D5 → TILT DIR | |
| ☐ W10 5 V bus → PAN VDD | ☐ W20 Nano D7 → laser S | |

Power ON / OFF order

- | | |
|-----|------------------------|
| ON | 1. USB into the laptop |
| ON | 2. switch 12 V ON |
| OFF | 1. switch 12 V OFF |
| OFF | 2. USB out |

Stop with the switch at any heat, smell or grinding sound.

Set Vref = 0.55 V with USB only, 12 V OFF.
Heat sink on every A4988 chip.
AS5600 sensor: separate sheet, later.